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|----|-----|--|-------------|---|
| 1. | (a) | $(1.5 \times 1000 \text{ kg}) \times 4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \times (80 - 60) \text{ }^{\circ}\text{C} \times (1-15\%)$
$= 1.07 \times 10^8 \text{ J}$ | 1M+1M
1A | 3 |
| | (b) | $1.07 \times 10^8 \text{ J} \div (4.5 \text{ kW}) \div 3600 \text{ s}$
$= 6.61 \text{ (hours)}$ | 1M
1A | 2 |
| | (c) | Rate of heat transfer drops as the water temperature drops / the room temperature increases / temperature difference drops. | 1A | 1 |

2. (a) Stop heating and stir the water well.

1A
1A 2

(b) (i)
$$\frac{L - 64}{80 - 64} = \frac{65 - 20}{92 - 20}$$
$$L - 64 = 10$$
$$L = 74 \text{ mm}$$

1M

1A 2

(ii) Let x °C be the absolute zero.

$$\frac{20 - x}{92 - 20} = \frac{64 - 0}{80 - 64}$$
$$20 - x = 288$$
$$x = -268 \text{ °C}$$

1M

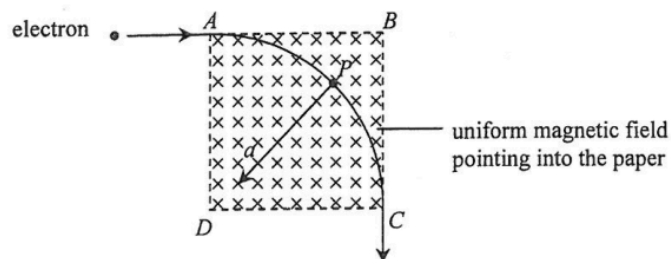
1A 2

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|-----|-------|------|---|----|----------|
| 3. | (a) | (i) | $P = F v$ | 1M | |
| | | | $= 8000 \text{ N} \times 2 \text{ m s}^{-1}$ | 1A | <u>2</u> |
| | | | $= 16 \text{ kW}$ | | |
| | | (ii) | $P_{\text{loss}} = 20 \text{ kW} - 16 \text{ kW} = 4 \text{ kW}$ | 1A | <u>1</u> |
| (b) | (i) | | $P = 4 \text{ kW} + (8000 - 7000 \text{ N}) \times 2 \text{ m s}^{-1}$ | 1M | |
| | | | $= 6000 \text{ W} = 6 \text{ kW}$ | 1A | <u>2</u> |
| | (ii) | | Output power required from motor is smaller. | 1A | |
| | | | <u>Or</u> Force exerted by motor is smaller. | | <u>1</u> |
| | (iii) | | No, the lift system could not work / the lift will fall | 1A | |
| | | | as slipping would occur / the cable cannot be fixed on the drum / the drum cannot exert a | 1A | |
| | | | force on the cable. | | <u>2</u> |

4. (a) (i) $F = qvB = (1.60 \times 10^{-19} \text{ C})(1.2 \times 10^7 \text{ m s}^{-1})(0.01 \text{ T})$
 $= 1.92 \times 10^{-14} \text{ N}$

1M
1A 2

(ii)



1A 1

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4. (b) F is perpendicular to the electron's displacement / velocity,
Or electron only changes direction while speed remains unchanged,
 no work is done, therefore k.e. remains unchanged.

1A
1A 2

(c)
$$\left. \begin{aligned} F &= qvB = \frac{mv^2}{r} \\ \frac{v}{r} &= \frac{qB}{m} = \text{constant} \end{aligned} \right\}$$

1M

v should be halved, i.e. $0.6 \times 10^7 \text{ m s}^{-1}$

1A 2

5. Stick the paper strip onto the horizontal part BC of the track. 1A
 Release the toy from a certain height h from the bench surface and measure the corresponding stopping distance d . 1A
 Stopping distance should be measured from the beginning of the horizontal part BC or on the paper strip. 1A
 Release the toy from different heights and measure the corresponding stopping distances. 1A
 Plot a graph of d against h , 1A
 a straight line passing through the origin should be obtained. 1A

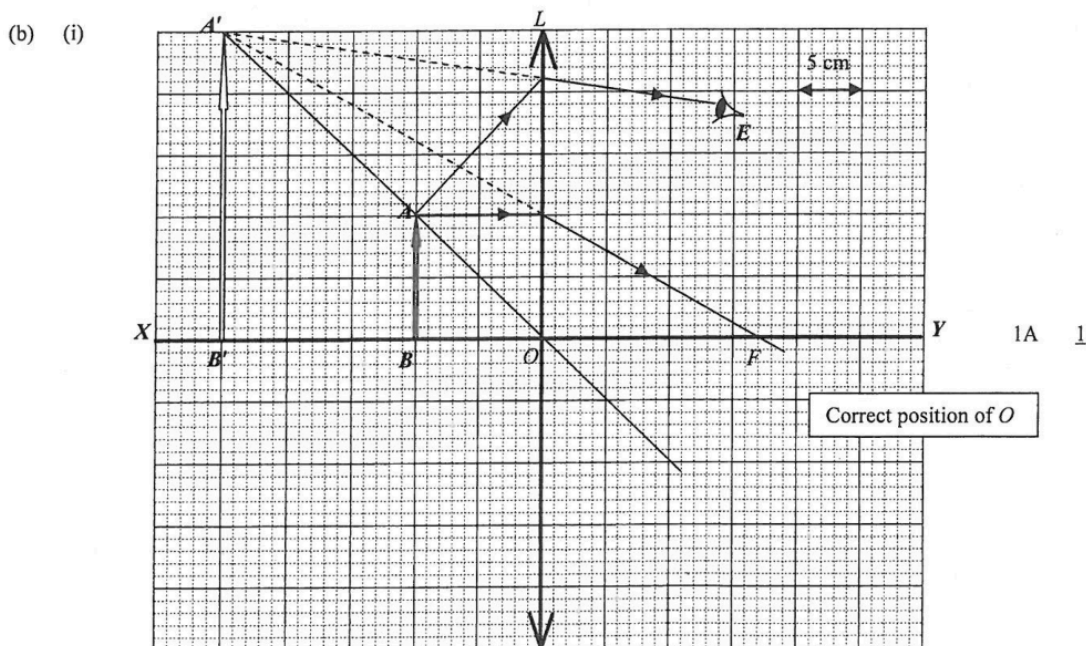
or

since $mgh = Fd$ so $\frac{d}{h} = \text{constant} / d \propto h$
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|----|-----|------|---|----|----------|
| 6. | (a) | (i) | Accelerates at g before the elastic cord stretches / at the beginning. | 1A | |
| | | | Acceleration decreases as the cord stretches. | 1A | |
| | | | Decelerates until momentarily at rest | 1A | |
| | | | (after the tension in the cord is greater than mg). | | <u>3</u> |
| | | (ii) | Gravitational potential energy changed to kinetic energy and | 1A | |
| | | | (then) elastic potential energy in elastic cord. | 1A | <u>2</u> |
| | (b) | | Elastic cord lengthens the stopping time, | 1A | |
| | | | hence reduces the (net) force acting on the player. | 1A | <u>2</u> |
| | (c) | | Contact area is larger, | 1A | |
| | | | hence pressure is smaller during the fall and the structure is less likely to break / detach. | 1A | <u>2</u> |

7. (a) $c = f\lambda \Rightarrow 3 \times 10^8 \text{ m s}^{-1} = f(0.02 \text{ m})$ 1M
 $\therefore f = 1.5 \times 10^{10} \text{ Hz or } 15000 \text{ MHz}$ 1A 2
- (b) (i) Path difference of the diffracted waves from slits A and B to probe varies along XY .
Constructive and destructive interference occur alternately to give maxima and minima. 1A 2
- (ii) $BP - AP = 1\frac{1}{2}\lambda$ 1M
 $BP - AP = 3 \text{ cm} = 0.03 \text{ m}$
 $\therefore BP = 1.24 + 0.03 = 1.27 \text{ m}$ 1A 2
- (iii) Path difference along $XY < AB$ 1M
 $AB = 3 \times 2 \text{ cm} = 3\lambda$
 $\therefore$ path difference allowed $= 0\lambda, 1\lambda, 2\lambda$.
Maximum number of maxima $= 3$ 1A 2
- (c) Radio waves with lower frequencies (will have longer wavelengths and hence) have greater diffraction effect. 1A
Radio waves by-pass small obstacles / not to be reflected from small obstacles. 1A 2

8. (a) (i) Virtual 1A 1
(ii) Convex.
Only convex lens can form magnified (virtual, erect) images. 1A 2

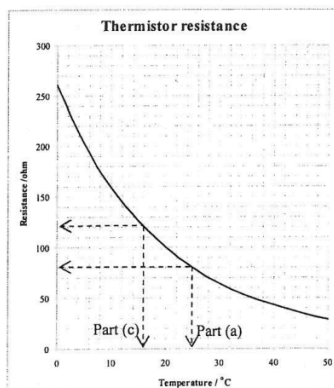


- (ii) Correct light ray to locate F . 1M
Focal length $f = 17 \text{ cm}$ (16.0 to 17.5 cm) 1A 2
- (c) Correct ray from A' or lens to E . 1A
All correct. 1A 2
- (d) Magnifying glass / glasses for long-sighted eyes / simple microscope 1A 1

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9. (a) $k = \frac{\ln 2}{5730 \times 3.16 \times 10^7} = 3.83 \times 10^{-12} (\text{s}^{-1})$ 1A
- Activity $A = kN$
 $N = \frac{A}{k} = \frac{0.2}{3.83 \times 10^{-12}}$ 1M
 $= 5.22 \times 10^{10}$ 1A 3
- (b) No. of ^{14}C nuclei: $N_0 = 1 \times 10^{23} \times (1.3 \times 10^{-12}) = 1.3 \times 10^{11}$ 1A 1
- (c) $kt = \ln \frac{N_0}{N}$
 $(3.83 \times 10^{-12}) t = \ln \frac{1.3 \times 10^{11}}{5.2 \times 10^{10}}$ 1M

10. (a) (i) 80Ω



1A 1

$$(ii) V_{AB} = \frac{120}{(80 + 120)} \times 12 = 7.2 \text{ V}$$

1M

1A 2

- (b) As R_v and 120Ω resistor are in parallel, R_{eq} across AB is smaller than 120Ω , therefore voltage shared across AB is reduced / smaller than expected.

1A

1A

Use a voltmeter with resistance much larger than the resistance in that part of the circuit. (e.g. $10 \text{ M}\Omega$ in some digital voltmeter)

1A

3

$$(c) (i) V_{AB} = \frac{120}{(R + 120)} \times 12 = 6.0 \text{ V}$$

$$R = 120 \Omega$$

1A

1A 2

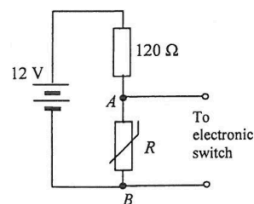
corresponds to temperature at 16°C .

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10. (c) (ii) Correct circuit (i.e. interchange thermistor R and 120Ω resistor).

As the temperature drops, the thermistor resistance increases.

When the resistance increases to a value such that $V_{AB} = 6.0 \text{ V}$ or above, the electronic switch is on and it turns on the heating device.



1A

1A

1A 3

11. (a)

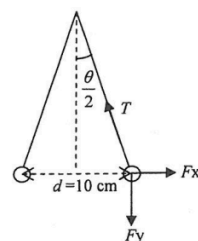
$$T \cos \frac{\theta}{2} = F_y = mg$$

$$T \sin \frac{\theta}{2} = F_x = \frac{Q^2}{4\pi \epsilon_0 d^2}$$

$$\tan \frac{\theta}{2} = \frac{Q^2}{4\pi \epsilon_0 d^2} \left(\frac{1}{mg} \right)$$

$$= 9 \times 10^9 \times \frac{(3.1 \times 10^{-9})^2}{0.1^2} \times \frac{1}{(1.0 \times 10^{-5})(9.81)}$$

$$\frac{\theta}{2} = 5.0^\circ \text{ i.e. } \theta = 10.1^\circ$$

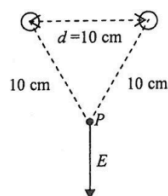


1M

1M

1A 3

- (b) (i)



1A 1

$$(ii) \text{ Potential at } P = \frac{Q}{4\pi \epsilon_0 d} + \frac{Q}{4\pi \epsilon_0 d} = \frac{2Q}{4\pi \epsilon_0 d}$$

1M

$$= (9 \times 10^9) \frac{2 \times 3.1 \times 10^{-9}}{0.1}$$

$$= 558 \text{ V}$$

1A 2

- (iii) Separation d decreases.

1A 1